

A Comparative Analysis of CNN Architectures for Audio Classification

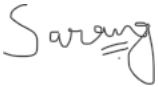
Sarang Bhatnagar

MSc in Statistics and Data Science

The University of Bath

September 2026

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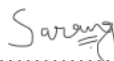
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Abstract

Audio classification is useful in many areas, including environmental monitoring and bird conservation. Large collections of audio recordings can be difficult and time-consuming to examine manually. Deep learning models can help by automatically identifying the most likely sound or species in each recording. However, larger and deeper models do not always provide better results, especially when the available training time and computing resources are limited.

This dissertation compares two convolutional neural network architectures, ResNet18 and ResNet50, for spectrogram-based audio classification. The main aim is to determine whether the additional depth and parameters of ResNet50 provide enough improvement to justify its greater computational cost. BirdCLEF 2024 was used as the main dataset, with experiments performed on the 10, 25, 30 and 50 most frequently represented bird species. ESC-50 was used as a secondary environmental-sound dataset.

All recordings were converted to five-second, 32 kHz mono audio and represented using mel spectrograms. Both models used ImageNet-pretrained weights and were fine-tuned using the same data split, preprocessing procedure, batch size, learning rate and five-epoch training schedule. Performance was compared using validation accuracy, macro-F1, weighted-F1, parameter count, inference time, bootstrap confidence intervals and paired McNemar tests.

ResNet18 achieved higher validation accuracy and F1 scores than ResNet50 in all four BirdCLEF experiments. However, the differences were small and were statistically significant only in the 50-class BirdCLEF experiment. In the primary 30-class experiment, ResNet18 achieved 81.41% validation accuracy compared with 80.86% for ResNet50. It also used approximately half as many parameters and was around 3.4 times faster during the measured inference procedure. The exploratory ESC-50 experiment also favoured ResNet18, although its random split contained recordings from shared sources and should not be treated as an official ESC-50 benchmark result.

Overall, ResNet18 provided the better balance between predictive performance and computational cost under the conditions tested. These findings do not show that ResNet18 is always better than ResNet50. Instead, they show that the additional capacity of ResNet50 did not provide an advantage under the limited five-epoch training protocol used in this study.

1. Introduction

Audio classification is the process of assigning a label to an audio recording or a segment of audio. It is used in applications including environmental monitoring, bioacoustics, industrial fault detection and urban sound analysis. In environmental research context, recording devices collect large quantities of sound without requiring a researcher to be present at each location, often called as passive acoustic monitoring.

Bird-species identification is an important application of passive acoustic monitoring. Bird populations can provide information about biodiversity and environmental change, but manually reviewing extensive the recordings is time consuming and requires domain knowledge. Therefore, automated classification systems can assist researchers by identifying recordings that are likely to contain particular species. Systems such as BirdNET show how deep learning models can support large scale bird monitoring, although performance of these systems is affected by recording conditions, background noise and differences between training recordings and natural soundscapes (Kahl et al., 2021).

This dissertation uses BirdCLEF 2024 as its primary dataset. The dataset focuses on under-studied bird species from the Western Ghats of India and contains recordings collected under varied environmental and recording conditions (Kahl et al., 2024). These conditions make bird-audio classification challenging. Recordings can contain background noise, differences in loudness and duration, overlapping vocalisations and variations within the calls of the same species. Different species may also produce acoustically similar vocalisations. Also, the number of available recordings is not equal for every species.

The official BirdCLEF task involves identifying potentially multiple species within continuous soundscapes. The experiments in this dissertation use a narrower scope based on selected primary labels from the open source training recordings. Each recording is assigned one target species, producing a **single-label multiclass classification problem**.

Audio recordings must be converted into a representation from which a classification model can learn. In this study, each recording is standardised to five seconds and transformed into a log-mel spectrogram. This represents acoustic energy across time and frequency and provides a two-dimensional input suitable for convolutional neural networks. The same preprocessing procedure is applied to every experiment so that both evaluated models receive inputs with identical dimensions and acoustic content.

Convolutional neural networks can learn local and hierarchical patterns from spectrograms, reducing dependence on manually selected acoustic features. However, increasing the size and depth of a CNN does not guarantee better performance. A larger model may learn more complex representations, but it also contains more parameters and requires additional training and inference computation. Whether this extra capacity produces a worthwhile improvement depends on the dataset, pretraining procedure and available optimisation budget.

This study compares ResNet18 and ResNet50, two residual CNN configurations with substantially different capacities. Both models are initialised using pretrained ImageNet weights and fine-tuned using the same selected recordings, mel-spectrogram representation, optimiser, learning rate, batch size and five-epoch schedule. ResNet50 has greater representational capacity, while ResNet18 requires fewer parameters and less computation. The practical question is therefore whether the larger model provides sufficient predictive improvement to justify its additional cost under a deliberately limited training protocol.

Four related BirdCLEF classification tasks are examined using the 10, 25, 30 and 50 most frequently represented primary labels. The 30-class task is treated as the primary experiment, while the other class sets provide additional evidence about the relative performance of the configurations. Expanding the class set also changes the species composition, number of recordings and class distribution. These experiments therefore do not isolate the causal effect of class count.

ESC-50 is used as a secondary environmental-sound dataset. It contains 2,000 five-second recordings distributed equally across 50 sound categories (Piczak, 2015). This experiment examines whether the relative pattern between ResNet18 and ResNet50 also appears outside bird-species classification. Because the experiment uses a random stratified split rather than the official source-aware folds, it is treated as exploratory supporting evidence.

1.1 Aim and Objectives

The aim of this study is to compare pretrained ResNet18 and ResNet50 configurations under a common, limited fine-tuning protocol and determine whether the additional capacity of ResNet50 provides a meaningful predictive advantage relative to its computational cost.

The objectives are:

1. To construct selected 10-, 25-, 30- and 50-class BirdCLEF classification tasks using the most frequently represented primary species labels.
2. To apply a consistent audio-preprocessing pipeline and mel-spectrogram representation to both architectures.
3. To fine-tune ResNet18 and ResNet50 using the same training and validation partitions and five-epoch training configuration.
4. To compare the models using accuracy, macro-F1, weighted-F1, per-class metrics, confusion matrices and paired prediction analysis.
5. To compare parameter count, training time and measured inference cost alongside predictive performance.
6. To conduct a secondary exploratory comparison using ESC-50.

1.2 Research Questions

The study addresses the following research questions:

RQ1: Under the same five epoch fine-tuning protocol, how do pretrained ResNet18 and ResNet50 compare in validation accuracy, macro-F1 and weighted-F1 on selected BirdCLEF 2024 primary-label classification tasks?

RQ2: Does the relative performance of the two pretrained configurations remain consistent across the selected 10-, 25-, 30- and 50-class BirdCLEF tasks?

RQ3: Does an exploratory comparison on ESC-50 provide supporting evidence that the observed architectural pattern extends beyond bird-species classification?

RQ4: What relationship exists between predictive performance and computational requirements, including parameter count and measured inference time, for ResNet18 and ResNet50 in the primary 30-class experiment?

This dissertation provides a controlled comparison of pretrained ResNet18 and ResNet50 for audio classification under a five-epoch fine-tuning protocol. It combines accuracy and F1 scores with paired error analysis, parameter counts and measured inference time to assess whether the larger model offers a practical benefit.

The study focuses on four selected BirdCLEF primary-label classification tasks, with ESC-50 used as an exploratory comparison. Each recording is assigned one target class. The findings apply to the datasets, data splits, pretrained weights, training settings and computing environment used in this study. They provide evidence for choosing between these configurations under the tested conditions.

2. Literature Review

Automatic audio classification has developed from traditional machine learning classifiers to deep-learning approaches which are capable of learning complex representations from time-frequency inputs. Convolutional neural networks are used widely for audio classification problems which take an image of a mel-spectrogram generated from the audio as input. However, model performance depends not only on the model architecture, but also on factors such as the available computational resources, model pretraining datasets, optimisation methods and many more variables.

The aim of this literature review is to find relevant sources that may potentially find any gaps in the research. In the following sections, an audio classification problem specific to environmental and bioacoustics is discussed because of the use of the BirdClef dataset in the study. We then discuss the audio representations and Convolutional Neural Network classifiers (CNNs).

2.1 Audio Classification in Environmental and Bioacoustic Contexts

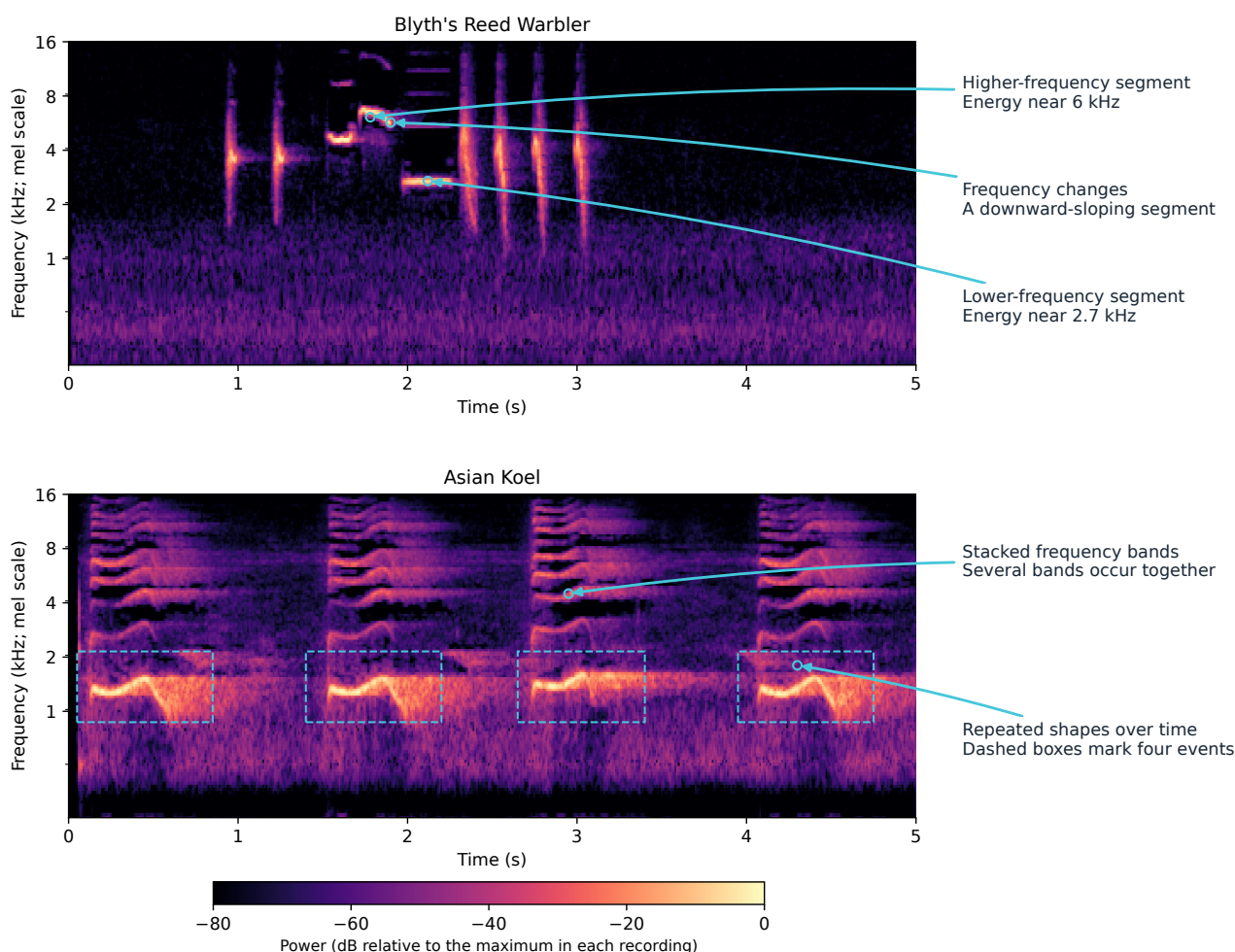
Audio classification involves assigning one or more labels to an audio recording, but the exact formulation of the task varies by domain. In environmental sound classification, datasets such as ESC-50 usually frame the task as single-label prediction across a diverse set of everyday sound categories, including animals, natural events, and human activities. Bioacoustic classification is more specialised, aiming to identify vocalisations within a narrower acoustic setting, such as distinguishing between bird species. This difference supports the use of BirdCLEF as the main dataset, while ESC-50 provides a secondary test of whether the architectural comparison transfers beyond one acoustic domain.

2.2 Audio Representations for Classification

Raw audio is a one-dimensional waveform containing amplitude measurements over time. Although this data preserves the original temporal signal, many important characteristics are related to frequency. For example bird vocalisations may differ in pitch, harmonic structure, duration and frequency modulation. These characteristics are difficult to observe directly from a one-dimensional waveform. Therefore this one-dimensional data is converted to two-dimensional time-frequency representation before classification. This representation is called a spectrogram.

Spectrograms can show repeated patterns, changes in frequency and several frequency bands occurring at the same time, as illustrated in Figure 2.2. These structures help describe how the sound changes over time.

Figure 2.2 Visible patterns in bird mel-spectrogram



A spectrogram is created by Short-Time Fourier Transform (STFT) to short, overlapping segments of a audio signal. This shows how the frequency content of the signal changes over time. The resulting representation contains time on the x-axis, and frequency on your axis while the color map shows the intensity of the signal. Spectrograms are useful for bird-audio classification because vocalisations can have clear visible structures such as repeated bird calls, pitch, harmonics and more. However, the quality of the spectrogram depends on settings such as the window length and the hop length.

The frequency axis of the obtained spectrogram is then converted into the Mel-scale, which gives more detail to lower frequencies and less detail to higher frequencies. The signal values are also converted to a logarithmic scale. This gives a two-dimensional representation (image) of the audio, which can be used as an input in the CNN models.

Mel frequency cepstral coefficients (MFCCs) are another way of representing audio. Instead of keeping the full time frequency representation they reduce the audio into a smaller number of features. Earlier bird classification systems used this with classifiers such as Support Vector Machines (SVMs). For example, Fagerlund (2007) used mel-cepstral features with SVMs for bird species recognition. These methods rely heavily on the choice of features selected before training.

In recent years, CNNs have changed how the audio is represented and used for classification. Instead of selecting a small set of features, spectrograms can be given directly to the CNNs and the filters of the model learn the hidden patterns in the image data. Piczak (2015) used this approach for environmental sound classification, and later studies also applied pretrained image classification models to audio spectrograms (Piczak, 2015; Palanisamy et al., 2020). This makes log-Mel spectrograms suitable for CNN-based classification and transfer learning which are also used during the experiments performed in this study.

However, Mel spectrograms are still based on predefined settings rather than learning directly from the data. Their output depends on things such as sampling rate, window size, hop length and number of Mel filters. Also they do not retain all the information from the original raw audio signal.

This study uses log-mel spectrograms because they are considered standard to be used in bioacoustic classification and can be supplied to both ResNet architectures in our study. Using the same representation for the comparison ensures that differences in performance evaluation is not caused by the differences in the input data. The exact preprocessing parameters and the procedure is used to convert each spectrogram into a three-channel model input as described in Section 3.2.

2.3 CNN-Based Audio Classification

Convolutional neural networks were originally developed for image processing tasks, but they can also be applied to two dimensional image representations such as spectrograms. When a spectrogram is supplied to a CNN, the network processes it as a two-dimensional array in which one axis represents time and the other represents frequency. Convolutional filters move across this image and learn local acoustic structures such as frequency bands and harmonics. During training the model learns which of these patterns are useful for classifying different classes.

A CNN contains many convolutional layers arranged sequentially. Filters in the earlier layers learn relatively simple local structures, while the filters in the later layers learn more complex patterns from the image by combining outputs of previous layers. For example, in bird audio classification, these higher-level patterns could be a combination of harmonics, changes in frequency, and other acoustic features. Pooling operations are used between convolutional layers to reduce the dimensions of feature maps and make the model less sensitive to small shifts in where a sound occurs in the spectrogram. The final learned representation is then passed to a classification layer, which produces an output vector with one value for each class. The softmax function is then applied to these final values to get the class probabilities.

An important advantage of convolution is parameter sharing. The same filter is applied across different parts of the spectrogram, which reduces the number of parameters and allows similar acoustic patterns to be detected at different times. However, time and frequency shifts do not have exactly the same meaning in a spectrogram. A sound occurring later in time may still represent the

same event, while moving a pattern along the frequency axis changes its pitch. Therefore, the benefits of convolution are not necessarily the same across both dimensions of the spectrogram.

Early research demonstrated that CNNs could learn useful representations directly from spectrogram-based inputs. Piczak (2015) evaluated a CNN containing two convolutional layers followed by fully connected layers on several environmental-sound datasets. The system operated on time frequency representations and achieved improved performance over the traditional baselines considered in that study. This provided evidence that convolutional models could learn relevant acoustic patterns without relying on handcrafted feature vectors. Nevertheless, the reported results were specific to the datasets, preprocessing choices and baseline methods used in the experiment, and should not be interpreted as demonstrating that CNNs are universally superior to all feature based classifiers ([Piczak, 2015](#)).

The effectiveness of a CNN also depends on the amount and diversity of training data. Environmental and bioacoustic datasets can be relatively small compared with the image datasets commonly used to train deep neural networks. A model with a large number of parameters may learn characteristics specific to the training recordings instead of patterns that generalise to unseen data. Background noise, recording equipment, location and recording conditions can also make the dataset hard to generalise. Salamon and Bello (2017) investigated this problem and found that audio data augmentation improved the performance of a deep CNN in their experimental setting. Their augmentation process altered training recordings while keeping the labels, increasing the variety of examples presented to the model. This work demonstrates that CNN performance cannot be attributed solely to architecture because the training and augmentation process can influence the resulting classifier ([Salamon and Bello, 2017](#)).

Hershey et al. (2017) compared several established CNN architectures, including AlexNet, VGG, Inception and residual networks, for large-scale audio classification, showing that vision-derived architectures transfer to audio but that performance depends on dataset size and class count as much as on architecture choice (Hershey et al., 2017).

CNNs also have some limitations for audio classification. Their performance remains dependent on the representation supplied to the network, meaning that information removed during preprocessing cannot subsequently be recovered by the model. Pooling and downsampling can improve computational efficiency and expand the effective receptive field, but excessive reduction may remove fine temporal or frequency information. Conventional convolutional layers initially model local regions, so learning relationships between acoustically distant events requires either multiple stacked layers, larger receptive fields or additional mechanisms for temporal modelling. Increasing network depth may help the model learn more complex representations, but it also increases parameter count, computational cost and optimisation difficulty. A deeper network therefore does not necessarily provide a proportional improvement in classification performance.

These considerations motivate the use of residual networks, which introduce shortcut connections to improve the optimisation of CNNs. Residual architectures have been applied a lot to spectrogram-based audio classification and transfer learning. However, the practical benefit of increasing residual-network depth remains dependent on the dataset, input representation, pretraining procedure and available training budget. The following section therefore examines residual learning, transfer from image classification and the relationship between model capacity and audio-classification performance.

2.4 Residual Networks, Transfer Learning and Model Capacity

Increasing the depth of a CNN allows the model to combine simple acoustic structures into more abstract features, but a deeper network does not automatically perform better. In practice, very deep networks can be harder to optimise because gradients have to pass through many layers. This can lead to cases where a deeper model performs worse than a shallower one, even when the problem is not simply overfitting. Residual networks were introduced by He et al. (2016) to address this issue. Instead of requiring each group of layers to learn a complete transformation from scratch, residual blocks learn a residual function which is added to a shortcut connection. This gives gradients a more direct path through the network and makes deeper architectures easier to train.

ResNet18 and ResNet50 are both residual networks, but they differ substantially in their block design and capacity, as described in Section 3.3. ResNet50 uses bottleneck blocks and has roughly twice as many parameters as ResNet18, meaning that the comparison is not a controlled test of depth alone. It is instead a comparison between two different residual configurations. In principle, ResNet50's greater capacity may help it learn more complex acoustic patterns, especially if there is enough data and training time. However, with a limited dataset, this extra capacity may also make the model more likely to learn recording-specific features, such as background noise, microphone characteristics, or other dataset-specific patterns that do not generalise well.

The literature also suggests that the relationship between depth and performance is not consistently monotonic. For example, Kong et al. (2020) found that, among their PANNs residual models, ResNet38 performed better than both the shallower ResNet22 and the deeper ResNet54 on AudioSet. This goes against a simple assumption that increasing depth always improves audio-classification performance. One possible reason is that depth also changes the network's receptive field. Koutini et al. (2021) showed that explicitly controlling receptive-field size improved performance across several audio tasks, suggesting that architecture-task fit may matter more than depth alone. This provides a plausible, although untested, explanation for the present comparison.

Transfer learning provides another way to improve performance without training a CNN entirely from random initialisation. In this approach, the model starts from parameters learned on another dataset and is then fine-tuned on the target task. ImageNet-pretrained CNNs have often been adapted for spectrogram-based audio classification. Although spectrograms represent time and frequency rather than normal image space, early convolutional filters trained on images can still capture general local structures such as edges, textures, and simple patterns. Palanisamy et al. (2020) showed that ImageNet-pretrained CNNs can provide strong baselines for audio classification and may converge in fewer epochs than models trained from scratch, although they did not directly compare ResNet18 and ResNet50.

A more relevant comparison is provided by Theivaprakasham et al. (2023), who evaluated ImageNet-pretrained ResNet18 and ResNet50 for mel-spectrogram bird-audio classification. They reported 89% accuracy for ResNet18 and 91% for ResNet50, which suggests that ResNet50 can outperform ResNet18 on a similar task. However, their training protocol first trained only the classification head for five epochs with the backbone frozen, before fine-tuning the full network for a further 50 epochs. This is a much longer optimisation budget than the five epochs used in this dissertation. Therefore, ResNet50's advantage in their study may depend partly on having enough training time for its additional capacity to become useful. Under a shorter training budget, a smaller model such as ResNet18 may adapt more quickly, although longer training can also increase the risk of overfitting.

Overall, the literature does not support a general claim that deeper residual models always outperform shallower ones. For this reason, this dissertation evaluates ResNet18 and ResNet50 as two pretrained residual configurations under the same short fine tuning protocol.

2.5 Alternative Audio Architectures and Computational Efficiency

CNNs mainly learn from small local areas of a spectrogram, so they may find it difficult to connect sound patterns that occur far apart in time. Adding more layers can help increase the amount of context the network captures, but this also increases computational cost. CRNNs and transformers approach this problem in different ways. A CRNN first uses convolutional layers to extract features, then uses a recurrent layer to model how those features change over time. Transformers instead use self-attention, which allows different parts of the input to be compared directly without using recurrence.

Cakir et al. (2017) showed that CRNNs work well for sound-event detection, where the model needs to detect when different sounds start, end, and overlap. However, that task is different from this dissertation, which focuses on single-label classification from a fixed set of classes. Therefore, their study suggests that recurrence may be useful, but it does not prove that it would improve the results here. Similarly, Gong et al. (2021) showed strong results with the Audio Spectrogram Transformer, but their model relied heavily on large-scale pretraining and data augmentation. These conditions are not the same as the short five-epoch fine-tuning setup used in this study.

Efficiency is also important when comparing models. Kong et al. (2020) and Schmid et al. (2024) show that making a model deeper or more complex does not always lead to a proportional improvement in performance. In some cases, a small gain in accuracy may not be worth the much larger number of parameters or higher inference cost. This trade-off is examined directly in Section 4.3 of this dissertation.

For this reason, comparing ResNet18 and ResNet50 remains relevant. Both models are widely used, easy to reproduce, and based on the same residual-network design, but they differ clearly in size and capacity. This makes them suitable for testing whether the extra capacity of ResNet50 is worth the additional computational cost under a limited fine-tuning budget. The aim of this dissertation is not to find the best possible audio-classification architecture overall, but to compare two common residual networks in a clear and reproducible way. However, reproducibility depends on more than using the same implementation: the data split, random variation during training, and checkpoint selection can also affect the results, which are discussed in the next section.

2.6 Reliability and Fairness in Model Evaluation

A reliable comparison between two architectures cannot be based only on training both models on the same dataset. The data split, preprocessing steps, training settings, and evaluation method also need to stay the same. Otherwise, any difference in performance may come from the experimental setup rather than from the models themselves. Even when these factors are controlled, there are still some sources of uncertainty that should be acknowledged.

One source of uncertainty is training randomness. Bouthillier et al. (2021) showed that small choices such as parameter initialisation or minibatch order can change the reported results and even the ranking of methods. This means that a single-seed result represents only one possible training run, not the full range of outcomes the model might produce. Another issue is data splitting. In audio datasets, multiple clips may come from the same original recording. If these clips are split across training and validation sets, the model may benefit from similar background noise or recording conditions rather than learning features that generalise well. The ESC-50 dataset addresses this using five source-aware folds (Piczak, 2015). However, the exploratory ESC-50 experiment in this dissertation uses a random stratified split, so its results should be interpreted with this limitation in mind.

The way models are compared statistically is also important. Since ResNet18 and ResNet50 are tested on the same validation recordings, their predictions are paired. For this reason, McNemar’s test is more suitable than treating their accuracies as independent results (Dietterich, 1998). However, if this test is not significant, it does not prove that the two models are equivalent. It only means that this experiment did not find enough evidence of a difference. Checkpoint selection is another limitation. If the same validation set is used to choose the best epoch and report the final score, the reported performance can be slightly optimistic, because the model-selection process can overfit the validation criterion (Cawley and Talbot, 2010). Therefore, the results in this dissertation should be treated as validation results, not as fully unbiased test-set estimates, as discussed in Section 3.5.

Overall, controlling the experiment makes the comparison fairer, but it does not make the results fully generalisable. This dissertation keeps preprocessing, dataset, and training schedule consistent across both models. However, the findings are still limited by single-seed training, validation-based checkpoint selection, and possible source overlap in the ESC-50 split. These limitations define how far the results can be interpreted and are considered again in the final synthesis.

2.7 Research Gap

The reviewed literature shows a transition from traditional audio classification systems based on manually selected acoustic features to deep learning approaches. Earlier bird-recognition systems successfully combined features such as MFCCs with classifiers including support vector machines. Later research showed that CNNs could learn acoustic patterns directly from spectrogram representations. This mitigated the need of handcrafting features.

Residual connections made it easier to train deeper CNNs, so they are now widely used in audio classification. Transfer learning has enabled residual networks trained on large image or audio datasets to be fine tuned for smaller audio classification tasks saving computation and training time. Studies including Palanisamy et al. (2020) and Kong et al. (2020) demonstrate the effectiveness of pretrained CNNs for audio classification. However, these studies also indicate that performance is determined by the combination of architecture, pretraining, input representation and optimisation process rather than model depth alone (Palanisamy et al., 2020; Kong et al., 2020).

Previous literature has already compared residual architectures of different capacities. Hershey et al. (2017) compared several CNN families for large-scale audio classification, while Kong et al. (2020) evaluated audio-specific residual networks of different depths and reported their computational requirements. The PANNs results did not show a consistent improvement as residual-network depth increased. Therefore, a broad claim that the relationship between CNN architecture and audio-classification performance has not previously been investigated would not be supported by the literature (Hershey et al., 2017).

The exact ResNet18 and ResNet50 model pair has also previously been evaluated for bird-audio identification. Theivaprakasham et al. (2023) reported higher performance from ResNet50 using mel-spectrogram inputs. However, their models were trained using a substantially longer transfer-learning procedure consisting of five epochs of classifier-head training followed by 50 epochs of complete fine-tuning. Their dataset, segmentation procedure and preprocessing configuration also differed from those used in the present study. Their findings demonstrate that ResNet50 can provide an advantage under their experimental conditions, but they do not determine whether that advantage remains under a much shorter fine-tuning budget or on selected BirdCLEF 2024 classes (Theivaprakasham et al., 2023).

To clarify how the present study relates to previous research, Table 2.7 summarises the most relevant literature, its contribution and its limitations in relation to this study.

Table 2.7 Literature Review Summary

Study	Task and approach	Relevance and limitation
Fagerlund (2007)	Bird recognition using mel-cepstral and low-level features with SVMs	Demonstrates the traditional feature-based approach preceding deep spectrogram models.
Piczak (2015)	Spectrogram CNN for environmental-sound classification	Early evidence for CNN-based acoustic feature learning; based on comparatively small environmental datasets.
Hershey et al. (2017)	Comparison of several CNN architectures for large-scale audio classification	Shows that architectural comparisons already exist, but uses a much larger and different audio setting.

Palanisamy et al. (2020)	ImageNet-pretrained CNNs transferred to audio datasets	Supports image-to-audio transfer learning; does not provide a controlled ResNet18/ResNet50 short-budget comparison.
Kong et al. (2020)	Audio-pretrained CNNs, residual variants and computational analysis	Shows that greater residual depth does not consistently improve performance; uses audio-specific pretraining and different residual architectures.
Theivaprakasham et al. (2023)	ImageNet-pretrained ResNet18 and ResNet50 for bird-audio identification	Closest model-pair comparison; uses different data and a substantially longer 55-epoch transfer-learning procedure.
Schmid et al. (2024)	Efficient pretrained CNNs for audio classification	Establishes the importance of performance-complexity trade-offs; does not examine the specific models or hardware.

Table 2.7 shows that previous research has examined traditional audio classifiers, spectrogram-based CNNs, transfer learning, residual-network depth and computational efficiency. However, differences in datasets, pretraining sources and optimisation procedures mean that these findings do not directly determine whether ResNet50 provides a worthwhile advantage under the five-epoch protocol used in this study.

This distinction is important because a larger model may require more extensive optimisation before its additional capacity produces a clear benefit. ResNet50 also has higher parameter, memory, and computational demands, so a modest improvement in performance may not be sufficient to justify its extra cost. Conversely, if ResNet18 achieves comparable results, it may represent the more practical architecture. For this reason, predictive performance and computational efficiency need to be evaluated together.

Although existing literature provides strong evidence on high-performing architectures and large-scale pretraining, there is less evidence on the behaviour of readily available ImageNet-pretrained residual networks when they are fine-tuned under a short, resource-constrained training schedule. Most published comparisons use longer optimisation procedures. The gap addressed here is therefore practical and context-specific, rather than a claim that ResNet comparisons are absent from prior work.

This dissertation investigates that gap by comparing pretrained ResNet18 and ResNet50 under the same five-epoch training protocol across four BirdCLEF class-set tasks: 10, 25, 30, and 50 classes, as described in Section 3.1. It also includes an exploratory ESC-50 comparison to examine whether the relative performance pattern extends beyond bird-species classification. However, because each expanded class set also changes the species included, the number of recordings, and the class distribution, these tasks provide related but non-causal evidence about class-set effects, rather than isolating class count as the only variable.

In addition to accuracy and F1-based metrics, the study reports paired prediction outcomes, parameter counts, and inference time. This is because any predictive advantage must be considered alongside its computational cost before it can be judged practically meaningful. Taken together, this evidence provides the basis for the research questions introduced in Section 1.2.

3. Methodology

3.1 Dataset and Class Selection

Two publicly available audio datasets were used in this study: BirdCLEF 2024 and ESC-50. BirdCLEF was selected as the primary dataset because it provides a challenging bioacoustic classification problem involving recordings collected under different environmental conditions. ESC-50 was used as a secondary dataset to test whether the relative performance of the two residual-network configurations was limited to bird-species classification or also appeared in general environmental-sound classification. Using publicly available datasets also supports the reproducibility of the experiments.

3.1.1 BirdCLEF 2024

BirdCLEF 2024 focuses on bird species from the Western Ghats of India. The released training dataset contains 24,459 focal recordings representing 182 primary species labels. The recordings were largely sourced from Xeno-canto and vary in duration, background noise, recording equipment and geographical location. This variation makes the dataset have practical difficulties encountered in bioacoustic classification (Kahl et al., 2024).

The official BirdCLEF challenge evaluates predictions on continuous soundscape recordings, which may contain multiple species and background events. However, the present study did not use the hidden competition test soundscapes. Only `train_audio` and `train_metadata.csv` were used. The results reported in this dissertation are therefore validation results obtained from selected focal training recordings and should not be interpreted as BirdCLEF competition scores.

Each recording in `train_metadata.csv` contains a `primary_label` identifying the principal recorded species. This field was used as the target class. Although the metadata can contain secondary species information, secondary labels were not included in the classification target. The experiments therefore formulate the data as a closed-set, single-label, multiclass classification problem. This formulation does not necessarily imply that no additional species are audible in a recording.

3.1.2 BirdCLEF Class-Subset Construction

Training on all 182 BirdCLEF classes was not practical due to computational limitations and time constraints. Four subsets containing 10, 25, 30 and 50 classes were therefore constructed. The classes were ranked according to their frequency in the `primary_label` column, and the most frequently represented classes were selected for each experiment. All recordings belonging to the selected classes were retained.

This procedure ensured that the selected species had comparatively strong representation in the training data. The 10-class subset was completely balanced because each selected species contained 500 recordings. The larger subsets were not completely balanced because class frequencies decreased as additional species were included. Consequently, these experiments should not be described as four balanced datasets.

The class sets were nested according to the frequency ranking. For example, the 10 species in the first experiment were also present in the 25-, 30- and 50-class sets. However, increasing the number of classes also changed the total number of recordings, the class distribution and the species included. The experiments therefore represent four related classification tasks rather than a controlled test in which only class count changed.

Number of classes	Total recordings	Training recordings	Validation recordings	Recordings per class
10	5,000	4,000	1,000	500
25	12,127	9,701	2,426	302-500
30	13,503	10,802	2,701	273-500
50	17,590	14,072	3,518	155-500

After filtering the metadata for each selected class set, the recordings were divided into training and validation partitions using an 80:20 split. Stratification was performed using the primary species label so that both partitions retained approximately the same class distribution. Random seed 42 was used to make the partitions reproducible.

Within each class-set experiment, ResNet18 and ResNet50 were trained and evaluated using the same selected recordings and split configuration. This enabled paired comparison of their predictions on the same validation examples. The 30-class subset was designated as the primary BirdCLEF experiment, while the 10-, 25- and 50-class subsets provided additional comparisons across related label sets.

3.1.3 ESC-50

ESC-50 is a benchmark dataset for environmental-sound classification introduced by Piczak (2015). It contains 2,000 five-second recordings distributed equally across 50 classes, with 40 recordings per class. The classes cover five broad categories: animal sounds, natural soundscapes and water sounds, human non-speech sounds, interior and domestic sounds, and exterior and urban noises. Unlike BirdCLEF, ESC-50 contains a broad range of environmental events rather than different species from the same biological group.

All 2,000 ESC-50 recordings were used. The same 80:20 stratified splitting procedure and random seed of 42 were applied, producing 1,600 training recordings and 400 validation recordings. Because ESC-50 is balanced, this resulted in 32 training and eight validation recordings per class. Both residual-network configurations were trained using the same split and the same audio-preprocessing procedure used for BirdCLEF.

ESC-50 provides five official folds designed to keep clips derived from the same original source recording together. The experiment in this study instead used a random stratified split. Inspection of the split metadata identified 129 original source identifiers appearing in both the training and validation partitions. The split therefore contained source-recording overlap, even though no individual clip appeared in both partitions.

For this reason, the ESC-50 experiment is treated as an exploratory secondary comparison rather than an official benchmark evaluation. Its purpose is to compare ResNet18 and ResNet50 under the same local protocol on a different type of audio data. The resulting scores should not be compared directly with published ESC-50 results obtained using the official source-aware five-fold procedure.

The audio standardisation and mel-spectrogram procedures applied to both datasets are described in Section 3.2.

3.2 Audio Preprocessing

The same preprocessing pipeline was applied to the BirdCLEF and ESC-50 recordings. This ensured that ResNet18 and ResNet50 received identical input representations within each experiment. The pipeline consisted of audio loading and duration standardisation, mel-spectrogram generation, decibel conversion, normalisation and conversion into a three-channel tensor.

3.2.1 Audio Loading and Standardisation

Audio files were loaded using the `librosa` library and converted to mono. All recordings were resampled to a target sampling rate of 32 kHz. A common sampling rate was required because the source recordings could have been recorded using different sampling frequencies. It also ensured that the spectrogram parameters represented the same time and frequency intervals across all recordings.

Each recording was standardised to a duration of five seconds, corresponding to 160,000 audio samples at 32 kHz. Recordings shorter than five seconds were padded with zeros at the end until they reached the required length. For recordings longer than five seconds, only the first five seconds were retained. The resulting waveform therefore had the same length for every input.

The five-second duration was selected to provide a consistent input size while keeping the computational requirements manageable. However, using the beginning of each recording means that vocalisations occurring later in a longer BirdCLEF recording were not included. This was a fixed preprocessing decision applied equally to both architectures rather than an attempt to identify the most informative segment of every recording.

ESC-50 recordings already have a duration of approximately five seconds. The same loading and resampling procedure was nevertheless applied so that ESC-50 and BirdCLEF were processed through a consistent pipeline.

3.2.2 Mel-Spectrogram Generation

Each standardised waveform was converted into a mel spectrogram using `librosa.feature.melspectrogram`. A Fourier transform size of 1,024 samples and a hop length of 320 samples were used. At a sampling rate of 32 kHz, the hop length corresponds to a temporal step of 10 milliseconds between consecutive frames.

The spectrogram contained 128 mel-frequency bands covering frequencies from 20 Hz to 16 kHz. The upper limit of 16 kHz corresponds to the Nyquist frequency for audio sampled at 32 kHz. These settings produced a power mel spectrogram with 128 frequency bins and 501 temporal frames for every five-second recording.

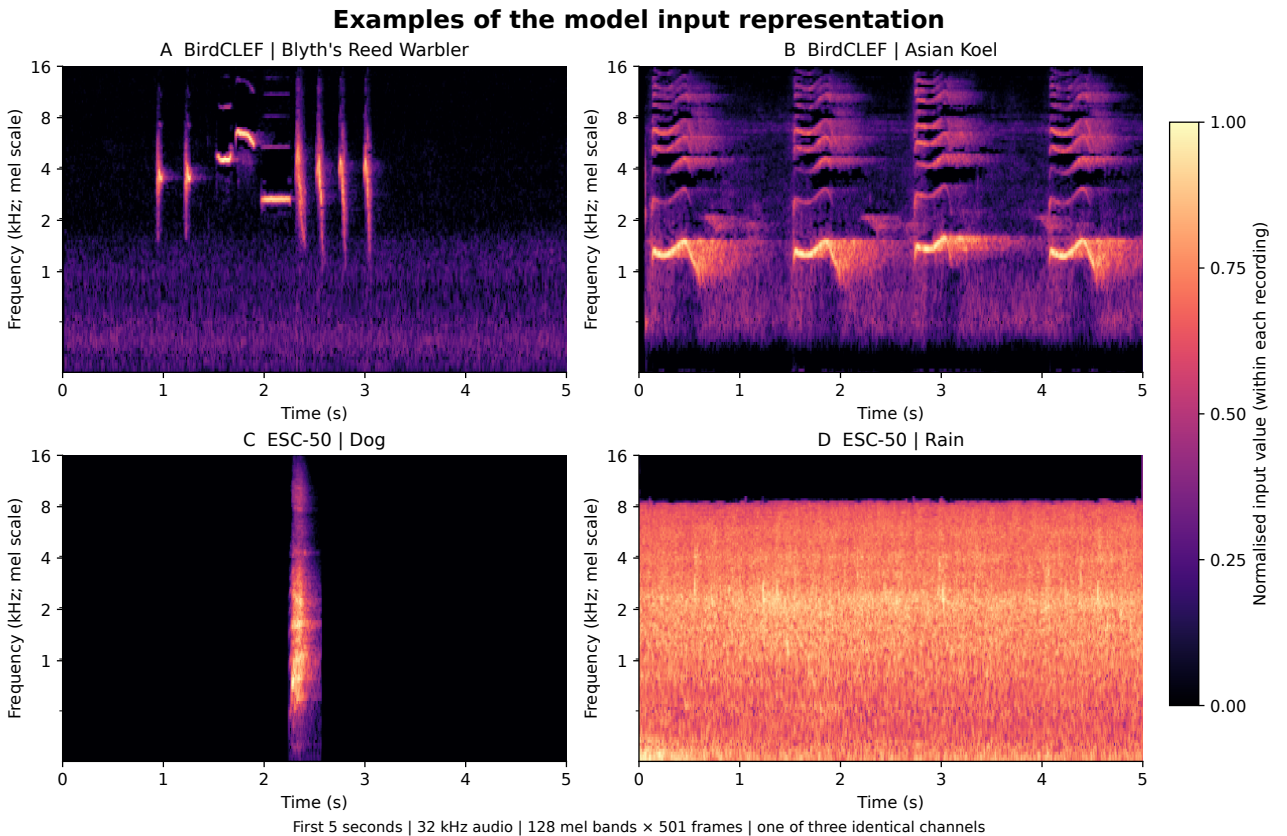
The spectrogram power values were converted to decibels using the maximum power within each recording as the reference value. As a result, the highest-power component in each spectrogram was represented as 0 dB, while lower-power components were represented using negative decibel values. This logarithmic conversion reduced the numerical range of the power values and made lower-energy acoustic structures more visible within the representation.

The preprocessing operation can be summarised as:

$$x[n] \rightarrow S(f, t) \rightarrow S_{\text{mel}}(m, t) \rightarrow S_{\text{dB}}(m, t) ,$$

where $x[n]$ represents the standardised waveform, $S(f, t)$ is its time-frequency power representation, $S_{\text{mel}}(m, t)$ is the mel-filtered power spectrogram and $S_{\text{dB}}(m, t)$ is the final decibel-scaled mel spectrogram.

Figure 3.1: Some examples of mel-spectrograms from the BirdCLEF and EAC-50 datasets



The visible structures in the spectrogram show how acoustic energy changes across time and frequency. Harmonic structures and frequency changes that may be difficult to identify directly from the waveform can be represented as patterns within the two-dimensional spectrogram. The figure is included to illustrate the model input rather than to provide an independent analysis of the selected species.

3.2.3 Normalisation and Model Input Preparation

After decibel conversion, each mel spectrogram was independently normalised to the range zero to one. For a decibel-scaled spectrogram S_{dB} , normalisation was performed using:

$$S_{\text{norm}} = \frac{S_{\text{dB}} - \min(S_{\text{dB}})}{\max(S_{\text{dB}}) - \min(S_{\text{dB}}) + \epsilon}$$

where $\epsilon = 10^{-8}$ was included to prevent division by zero. This procedure means that normalisation was calculated separately for every recording rather than from statistics calculated across the complete dataset.

The pretrained ResNet implementations used in this study expect an input containing three channels. A mel spectrogram contains only one channel, so the same normalised spectrogram was repeated three times. This produced a final input tensor with dimensions:

$3 \times 128 \times 501$, where the first dimension represents the three repeated channels, the second represents the 128 mel-frequency bands and the third represents the 501 temporal frames. The three channels contained identical values and therefore did not provide three different acoustic representations. Channel replication was used only to maintain compatibility with the pretrained ResNet input layer without modifying its expected number of channels.

The final spectrogram arrays were converted into 32-bit floating-point PyTorch tensors. Target species labels were mapped to sequential integer indices and represented as integer tensors for use with the multiclass classification loss. The mapping was generated separately for each BirdCLEF class subset, while ESC-50 used its provided numeric target labels.

The spectrograms were not resized to conventional square image dimensions. Both ResNet architectures received the complete $3 \times 128 \times 501$ tensor, preserving the full five-second temporal representation produced by the preprocessing pipeline. Because the same tensors were supplied to ResNet18 and ResNet50, differences between their results cannot be attributed to differences in the audio preprocessing or input dimensions.

3.3 Model Architectures

Two convolutional neural-network architectures were evaluated: ResNet18 and ResNet50. Both models were implemented using TorchVision and initialised with pretrained ImageNet weights. They received the same three-channel mel-spectrogram tensors described in Section 4.2. The purpose of selecting these models was to compare two widely available residual-network configurations with substantially different capacities while retaining a common architectural family.

Both models begin with a convolutional stem consisting of a 7×7 convolution, batch normalisation, a rectified linear unit activation and max pooling. This is followed by four stages of residual blocks. As the input progresses through the network, its temporal and frequency dimensions are gradually reduced while the number of feature channels increases. Adaptive average pooling is applied after the final residual stage, allowing both architectures to process the non-square $3 \times 128 \times 501$ spectrogram tensors without resizing them to conventional image dimensions.

The principal difference between the models is the construction and number of their residual blocks. Table 4.2 summarises the relevant architectural differences.

Table 4.2: Architectural comparison of the evaluated residual networks

Characteristic	ResNet18	ResNet50
Residual block type	Basic block	Bottleneck block
Blocks across four stages	2, 2, 2, 2	3, 4, 6, 3
Features before	512	2,048
Parameters in the 30-class model	11.19 million	23.57 million
Pretrained-weight request	<code>ResNet18_Weights.DEFAULT</code>	<code>ResNet50_Weights.DEFAULT</code>

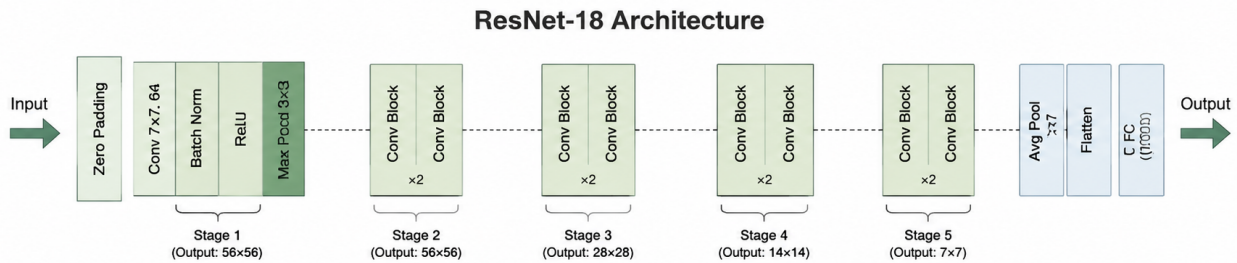
The parameter totals include the replacement classification layer used for the 30-class experiment. The exact total changes slightly between experiments because the number of output classes changes.

3.3.1 ResNet18

ResNet18 is the smaller of the two evaluated models. It uses basic residual blocks containing two 3×3 convolutional layers. A shortcut connection adds the input of each block to its transformed output. When the spatial dimensions or number of channels change, the shortcut path uses a projection to make the tensor dimensions compatible.

The four residual stages contain two basic blocks each. After the final stage, adaptive average pooling reduces each feature map to a single value. This produces a 512-dimensional representation for each input recording, which is supplied to the final classification layer.

For the primary 30-class experiment, the modified ResNet18 contained approximately 11.19 million trainable parameters. Its comparatively small parameter count and simpler residual blocks were expected to reduce training and inference cost relative to ResNet50. ResNet18 was therefore treated as the lower-capacity baseline in the comparison.

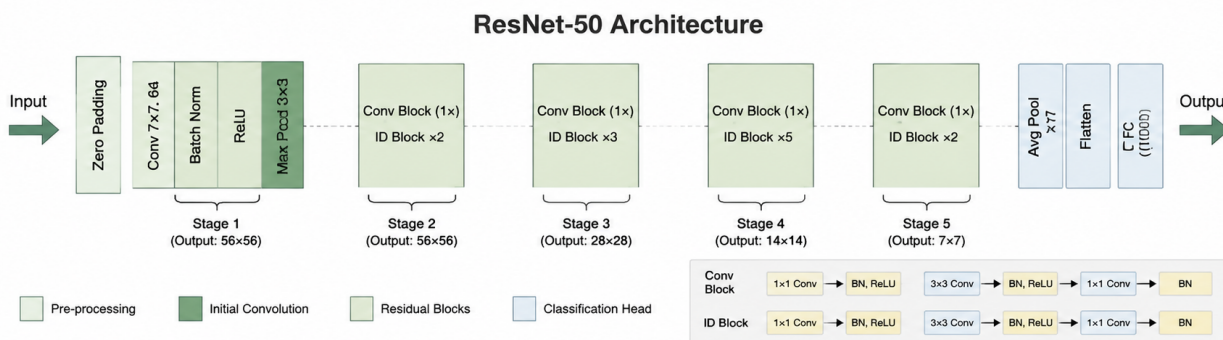


3.3.2 ResNet50

ResNet50 is a deeper residual architecture that uses bottleneck blocks. Each bottleneck block contains three convolutional operations. A 1×1 convolution first reduces the number of channels, a 3×3 convolution processes the resulting features, and another 1×1 convolution expands the channel dimension. As with ResNet18, a shortcut connection adds the block input to its output.

The four residual stages contain 3, 4, 6 and 3 bottleneck blocks respectively. The final stage produces a 2,048-dimensional representation before classification. For the primary 30-class experiment, the modified ResNet50 contained approximately 23.57 million trainable parameters. It therefore had approximately 2.1 times as many parameters as the corresponding ResNet18 model.

The greater depth and feature capacity of ResNet50 could allow it to learn more complex acoustic representations. However, it also requires more computation and memory. The experiment examined whether this additional capacity produced a meaningful predictive advantage under the same five-epoch training protocol.



3.3.3 Transfer Learning and Pretrained Weights

Model architectures described in Section 3.3 were initialised using weights learned from the ImageNet-1K image-classification dataset. The implementation requested the TorchVision DEFAULT pretrained-weight option for each model. In the installed environment, consisting of PyTorch 2.13.0 and TorchVision 0.28.0, these aliases resolved to `IMAGENET1K_V1` for ResNet18 and `IMAGENET1K_V2` for ResNet50.

Both models therefore used the same ImageNet source dataset, but they did not use identical pretraining recipes. The ResNet50 V2 weights were produced using a newer TorchVision training recipe than the ResNet18 V1 weights. Due to this difference it cannot be concluded that the network depth is the only changing variable.

The original ImageNet models expect three-channel natural images. Because each normalised mel spectrogram was repeated across three identical channels, the original first convolutional layer and its pretrained weights could be retained without modification. The pretrained filters were therefore applied to spectrogram patterns rather than natural-image pixels.

ImageNet and audio spectrograms belong to different domains, so the transferred weights cannot be assumed to provide an audio-specific representation. They instead provide a pretrained initialisation that can be adapted to the target audio data. The experiments compare the practical performance of these readily available pretrained configurations rather than comparing ImageNet pretraining against random initialisation or audio-specific pretraining.

3.3.4 Classification-Layer Modification

The original ImageNet classification layer produces outputs for 1,000 image classes. This layer was replaced with a new fully connected layer whose number of outputs matched the number of target classes in each experiment.

For ResNet18, the replacement layer received 512 input features. For ResNet50, it received 2,048 input features. The output size was set to 10, 25, 30 or 50 for the BirdCLEF experiments and 50 for ESC-50. Each output represented the unnormalised classification score, or logit, associated with one target class.

No softmax operation was included as a separate model layer during training because the selected cross-entropy loss function operates directly on logits. During evaluation, the class with the highest output score was selected as the model prediction. This produced one predicted class for every recording, consistent with the single-label task formulation described in Section 4.1.

The pretrained backbone was not frozen. All parameters in the residual blocks and the newly created classification layer remained trainable. Consequently, both architectures were fine-tuned end to end for the target audio-classification tasks rather than being used only as fixed feature extractors.

The same model-construction procedure was used across the BirdCLEF and ESC-50 experiments. Only the number of classification outputs changed according to the task. The optimiser, loss function, batch size, training duration and checkpoint-selection procedure are described in Section 4.4.

3.4 Training Setup

The same training configuration was used for ResNet18 and ResNet50 within every experiment. This included the optimiser, learning rate, batch size, loss function, number of epochs and dataset partition. Maintaining a common configuration allowed the pretrained model configurations to be compared under the same limited fine-tuning protocol.

All model parameters remained trainable. The pretrained residual layers and newly created classification layer were therefore updated together during training. The models were not trained using a frozen feature-extraction stage.

3.4.1 Optimisation Configuration

The models were trained using the Adam optimiser with a learning rate of 1×10^{-4} . No learning-rate scheduler or weight decay was applied. The optimiser was recreated separately for every model run.

Multiclass cross-entropy was used as the loss function. For a recording with target class y , the loss can be represented as:

$$L = -\log \left(\frac{\exp(z_y)}{\sum_{c=1}^C \exp(z_c)} \right),$$

is the model output for the correct class, z_c represents the output for class c , and C is the total number of classes. PyTorch’s cross-entropy implementation operates directly on model logits and internally applies the required normalisation.

No class weights were supplied to the loss function, and no weighted sampling procedure was used. Every training recording therefore contributed equally to the loss. This had no effect on class balance in the 10-class and ESC-50 experiments, but the larger BirdCLEF subsets contained unequal numbers of recordings across species.

The gradients were reset before every batch. A forward pass was performed to generate the model outputs, after which the loss was calculated and backpropagated through the complete network. The Adam optimiser then updated the model parameters. This procedure was repeated for each training batch.

The principal training hyperparameters are summarised in Table 4.3.

Table 3.3: Training configuration used across the experiments

Hyperparameter	Value
Optimiser	Adam
Learning rate	1×10^{-4}
Loss function	Multiclass cross-entropy
Batch size	32
Training epochs	5
Weight decay	None
Learning-rate scheduler	None
Data augmentation	None
Random seed	42
BirdCLEF data-loader workers	2
ESC-50 data-loader workers	0
Trainable parameters	Complete model

No data augmentation was applied, ensuring that both architectures were trained using the same unmodified recordings. Although augmentation methods can improve audio-classification performance when labelled data are limited (Salamon and Bello, 2017; Kahl et al., 2021), evaluating augmentation was outside the scope of this study. The results are therefore limited to the unaugmented training protocol.

3.4.2 Batch Construction and Training Duration

A batch size of 32 was used for both architectures and all class-set experiments. Training recordings were shuffled at the beginning of each epoch. Validation recordings were processed in a fixed order and were not shuffled.

Incomplete final training batches were discarded using `drop_last=True`. Consequently, the 10-class BirdCLEF and ESC-50 training sets used every recording during each epoch because their sizes were divisible by 32. In the other BirdCLEF experiments, a small number of recordings were omitted from each epoch’s final incomplete batch. Because the training data were shuffled, the omitted recordings could differ between epochs. No validation recordings were discarded.

Each model was trained for five epochs. This deliberately short budget was selected to keep the complete set of comparisons practical on the available hardware. No early-stopping rule was used to terminate training before the fifth epoch. Training and validation loss and accuracy were recorded after each epoch to produce the learning curves presented in Chapter 4.

Using the same number of epochs ensured that both architectures received the same number of passes through the available training data. However, this did not provide an equal computational budget. ResNet50 required more operations and more elapsed time per epoch than ResNet18. The five-epoch protocol should therefore be interpreted as a common training schedule rather than an equal-time comparison or evidence that either model had fully converged.

3.4.3 Random Seed and Reproducibility

Random seed 42 was used for NumPy and PyTorch. The same seed was also used when generating the stratified BirdCLEF training and validation partitions. This ensured that ResNet18 and ResNet50 were evaluated using the same recordings within each class-set experiment.

The same split and data-loader settings were used for both models, although the shared seed does not guarantee that every minibatch was presented in exactly the same order. Constructing models of different sizes can consume random state differently before data loading begins. Accelerator operations may also contain sources of nondeterminism. The experiments should therefore be described as single-seed comparisons on common partitions rather than fully deterministic replications.

Each experimental run saved a manifest containing the model name, selected classes, number of training and validation recordings, preprocessing configuration and principal hyperparameters. Training histories and summary metrics were also saved so that the reported results could be traced to their corresponding runs.

3.4.4 Validation and Checkpoint Selection

After every training epoch, the model was switched to evaluation mode and applied to the complete validation set without gradient calculation. Validation loss, accuracy, macro-F1 and weighted-F1 were recorded.

The checkpoint with the highest validation accuracy across the five epochs was retained. Only the model parameter state was saved; the Adam optimiser state was not included in the checkpoint. The selected model parameters were loaded after training, and the validation metrics, per-class metrics and confusion matrix were then generated from that checkpoint.

The same validation partition was used both to select the best epoch and to report the final validation performance. The resulting scores are therefore validation results rather than estimates from an untouched test set. This procedure is appropriate for comparing the saved models within the scope of this study, but it may produce optimistic estimates relative to evaluation on a completely independent test set. This limitation is considered when interpreting the results.

For most BirdCLEF experiments, the best checkpoint occurred at or near the fifth epoch. Nevertheless, selecting the checkpoint by validation accuracy means that the reported model was not necessarily the final epoch in every run. The selected epoch for each model is reported with the corresponding results in Chapter 5.

3.4.5 Computing Environment

The experiments were executed on a MacBook Air containing an Apple M3 processor and 16 GB of unified memory. PyTorch’s Metal Performance Shaders backend was used to perform training and inference on the Apple GPU. The saved experiment logs confirm that the BirdCLEF and ESC-50 runs used the MPS execution device.

The implementation automatically selected MPS when available, followed by CUDA and then CPU as fallback options. However, the results reported in this dissertation correspond to the MPS runs. The installed software environment used PyTorch 2.13.0 and TorchVision 0.28.0.

Wall-clock training time was measured from the beginning of the first training epoch to the completion of validation after the fifth epoch. It therefore included model optimisation, data loading and the validation pass performed after every epoch. The recorded training times are specific to the hardware, software environment and thermal conditions of the MacBook Air and should not be treated as hardware-independent values.

Training history, model parameters, validation predictions, per-class metrics and generated figures were saved after each experiment. The evaluation measures and statistical comparison procedures applied to these outputs are described in Section 4.5.

3.5 Evaluation Metrics and Comparison Procedure

The trained models were evaluated using predictive, statistical and computational measures. Predictive measures assessed the correctness of the classifications, while statistical analysis examined whether differences between the paired model predictions were greater than expected from the observed disagreement counts. Computational measures were used to compare model size and inference cost.

The best ResNet18 and ResNet50 checkpoints from each experiment were evaluated on the same ordered validation recordings. This ensured that the predictions could be compared directly at both the overall and individual-recording levels.

3.5.1 Classification Metrics

Accuracy was calculated as the proportion of validation recordings assigned to the correct class:

$$\text{Accuracy} = \frac{\text{Number of correct predictions}}{\text{Total number of predictions}}$$

Accuracy provides a clear overall measure of classification performance. However, it can be influenced by differences in class frequency because errors in more highly represented classes contribute more heavily to the final value. Additional class-based metrics were therefore calculated.

For each class c , precision was calculated as:

$$\text{Precision}_c = \frac{TP_c}{TP_c + FP_c}$$

where TP_c is the number of recordings correctly predicted as class c , and FP_c is the number incorrectly predicted as that class. Precision measures how often predictions for a particular class were correct.

Recall was calculated as:

$$\text{Recall}_c = \frac{TP_c}{TP_c + FN_c}$$

where FN_c is the number of recordings belonging to class c that were predicted as another class. Recall therefore measures how successfully the model identified the available examples of a class.

The F1-score was calculated as the harmonic mean of precision and recall:

$$F1_c = 2 \times \frac{\text{Precision}_c \times \text{Recall}_c}{\text{Precision}_c + \text{Recall}_c}$$

If a class contained no predicted positive examples, undefined metric values were assigned a value of zero.

Macro-F1 was calculated by averaging the individual class F1-scores while giving every class equal importance:

$$\text{Macro-F1} = \frac{1}{C} \sum_{c=1}^C F1_c$$

is the number of classes. Macro-F1 was particularly important for the 25-, 30- and 50-class BirdCLEF experiments because these subsets were not completely balanced. A model could only achieve a strong macro-F1 score by performing reasonably across both highly represented and less represented species.

Weighted-F1 was calculated by weighting each class F1-score according to the number of validation recordings belonging to that class:

$$\text{Weighted-F1} = \sum_{c=1}^C \frac{n_c}{N} F1_c$$

where n_c is the number of validation recordings from class c , and N is the total validation-set size. Weighted-F1 therefore reflects class frequency more strongly than macro-F1.

Accuracy, macro-F1 and weighted-F1 were reported as percentages. Per-class precision, recall, F1-score and support were also saved for diagnostic analysis.

3.5.2 Confusion Matrices

A confusion matrix was generated for each trained model. Rows represented the true classes and columns represented the predicted classes. Values on the main diagonal showed correct classifications, while off-diagonal values represented confusion between different species or sound categories.

The matrices contained raw prediction counts rather than percentages. They were used to determine whether an overall difference between the models was distributed across many classes or was primarily caused by a small number of classes. Per-class metrics were interpreted alongside the confusion matrices because two models with similar overall accuracy could still make different types of errors.

3.5.3 Bootstrap Confidence Intervals

A 95% bootstrap confidence interval was calculated for the accuracy of each model. The original correctness vector contained one value for every validation recording, indicating whether the model's prediction was correct or incorrect. From this vector, 1,000 bootstrap samples were generated by sampling validation examples with replacement.

Accuracy was calculated for each bootstrap sample. The 2.5th and 97.5th percentiles of the resulting accuracy distribution were used as the lower and upper limits of the 95% confidence interval. Random seed 42 was used for the bootstrap procedure.

The bootstrap intervals describe uncertainty associated with sampling from the observed validation recordings. They remain conditional on the trained checkpoint, selected validation split and training seed. They do not include variation that could occur if the models were retrained or evaluated using another data partition.

Confidence-interval overlap was considered descriptive information only. It was not used as the formal statistical test of the difference between ResNet18 and ResNet50 because the two models were evaluated on paired observations.

3.5.4 Paired McNemar Test

McNemar's test (McNemar, 1947) was used to compare the predictions of ResNet18 and ResNet50. This test was selected because both models were evaluated on exactly the same validation recordings. It examines cases where the models disagree in correctness rather than comparing their accuracy values as independent measurements.

Two discordant counts were calculated:

- b : recordings correctly classified by ResNet18 but incorrectly classified by ResNet50.
- c : recordings incorrectly classified by ResNet18 but correctly classified by ResNet50.

A continuity-corrected McNemar statistic was calculated as:

$$\chi^2 = \frac{(|b - c| - 1)^2}{b + c}$$

The statistic was compared with a chi-squared distribution with one degree of freedom. If no discordant predictions were present, the statistic was set to zero and the p-value to one.

A significance threshold of $p < 0.05$ was used. A significant result indicated evidence that the two saved classifiers had different probabilities of producing a correct prediction on the evaluated recordings. The direction of the difference was determined by comparing the discordant counts.

A non-significant result was not interpreted as proof that the models were equivalent. It indicated that the observed paired predictions did not provide sufficient evidence of a difference at the selected significance threshold. McNemar’s test also remained conditional on the particular trained checkpoints and validation partition. It did not account for variation caused by retraining the architectures using different random seeds.

The p-values were calculated separately for each experiment, and no multiple-comparison adjustment was applied in the implemented analysis. They should therefore be interpreted as experiment-specific tests rather than a single combined test across all datasets and class subsets.

Before McNemar’s test was calculated, the predictions from both models were aligned with the same ordered validation recordings. This ensured that each pair of predictions referred to the same sample.

3.5.5 Computational Measurements

The number of model parameters was calculated by summing the number of elements in all model parameter tensors. Because the complete networks were fine-tuned, the total parameter count was also the trainable parameter count. The size of each saved parameter checkpoint was recorded in megabytes.

Total training time was measured using wall-clock time from the beginning of the first training epoch until completion of validation after the fifth epoch. This measurement included the validation procedure performed after every epoch. It was not intended as a hardware-independent measure of algorithmic complexity.

Inference time was measured while each model processed the complete validation set in batches of 32. Before starting the timer for each batch, the input tensors were transferred to the MPS device and the accelerator was synchronised. The timer then measured the model forward pass. The

accelerator was synchronised again before the elapsed time was recorded to ensure that queued GPU operations had completed.

The accumulated forward-pass time was divided by the total number of validation recordings and converted to milliseconds per sample:

$$\text{Time per sample} = \frac{\text{Total timed forward-pass duration}}{\text{Number of validation recordings}} \times 1000$$

The measurement excluded audio loading, mel-spectrogram generation, transfer of the input tensors to the device and conversion of model outputs into predicted labels. It should therefore be described as **batch-amortised forward-pass time per sample**, rather than end-to-end deployment latency.

No separate inference warm-up stage or repeated benchmark rounds were implemented. The timing values represent a single pass through the validation loader and may be affected by runtime and hardware variation. They are suitable for comparing the two models within the recorded environment, but should not be treated as universal latency measurements.

The primary computational comparison was based on the 30-class BirdCLEF experiment. This allowed model size and inference cost to be considered alongside the predictive results obtained from the study’s primary class set.

3.5.6 Saved Evaluation Outputs

The comparison procedure reconstructed the same validation partition used during training and loaded the best saved checkpoint for each architecture. Predictions, true labels and timing measurements were then collected for the complete validation set.

The following outputs were retained:

- Overall accuracy, macro-F1 and weighted-F1.
- Bootstrap accuracy confidence intervals.
- McNemar statistics, p-values and discordant counts.
- Per-class precision, recall, F1-score and support.
- Confusion matrices containing raw prediction counts.
- Parameter counts and inference times.
- Epoch-level training and validation histories.
- Training and validation curves generated from those histories.

The saved outputs were used to produce the tables, figures and statistical comparisons presented in Chapter 5. BirdCLEF results are reported as validation results, while the ESC-50 comparison is explicitly treated as exploratory because its random split did not follow the official source-aware fold structure.

4. Experiments and Results

4.1 Overview

This chapter compares the performance of ResNet18 and ResNet50 on BirdCLEF and ESC-50. Both models followed the data preparation, preprocessing and training procedures described in Sections 3.1-3.4.

Four experiments were constructed using the 10, 25, 30 and 50 most frequent primary labels selected using the procedure described in Section 3.1.2. These class datasets were nested and therefore the smaller subset were also included in the larger subset. However, increasing the number of classes also increased the number of recordings. Table 4.1 summarises the experiments.

Performance was evaluated using accuracy, macro-F1, and weighted-F1 explained in Section 3.5.1. Predictions on same validation recordings were compared using McNemar’s test. Parameter counts, training times and inference times were examined isong the methods described in Section 3.5. The BirdCLEF results address RQ1 and RQ2, computational comparison addresses RQ2, and the exploratory ESC-50 comparison addresses RQ3. Section 4.5 summarises the findings and relates them to how they help answering the research questions.

Table 4.1: Experiments performed in the study

Dataset	Classes	Training recordings	Validation recordings	Role in the study
BirdCLEF-10	10	4,000	1,000	Small, balanced BirdCLEF subset
BirdCLEF-25	25	9,701	2,426	Larger, mildly imbalanced subset
BirdCLEF-30	30	10,802	2,701	Additional nested BirdCLEF subset
BirdCLEF-50	50	14,072	3,518	Broadest evaluated BirdCLEF subset
ESC-50	50	1,600	400	Exploratory secondary-dataset comparison

The Python code is available on GitHub’s public repository.

GitHub Link: <https://github.com/sarang997/dissertation>

4.2 BirdCLEF experiment

This section compares the models across the four BirdCLEF configurations. It finds overall validation performance, paired prediction outcomes, variation between species and training behaviour.

4.2.1 Overall Performance Across the Class Subsets

This section contains the results obtained by training ResNet18 and ResNet50 on the 10, 25, 30 and 50 classes of the dataset. The overall validation performance, difference in paired predictions, variation across individual species and learning behaviour.

Table 4.2 summarises the retained checkpoint results of each models accuracy, macro-F1 and weighted-F1 are reported because larger subsets contained unequal number of recording across classes (imbalance in dataset). The confidence intervals describe uncertainty associated with resampling the observed validation recordings, conditional on the trained model.

Table 4.2: Validation performance of ResNet18 and ResNet50 across the BirdCLEF class subsets

Classes	Model	Accuracy	Accuracy 95% CI	Macro-F1	Weighted-F1
10	ResNet18	86.80	[84.60, 88.80]	86.85	86.85
10	ResNet50	86.10	[83.90, 88.30]	86.07	86.07
25	ResNet18	85.74	[84.42, 87.14]	85.97	85.90
25	ResNet50	85.28	[83.88, 86.69]	85.37	85.35
30	ResNet18	81.41	[79.97, 82.86]	81.01	81.48
30	ResNet50	80.86	[79.45, 82.34]	80.60	81.08
50	ResNet18	78.82	[77.49, 80.16]	77.83	78.84
50	ResNet50	77.23	[75.92, 78.57]	75.81	77.27

ResNet18 achieved higher validation accuracy, macro-F1 and weighted-F1 scores in all the four experiments. The accuracy differences were approximately **0.70, 0.46, 0.55 and 1.59** percentage points for the 10, 25, 30 and 50-class comparisons respectively.

The difference between the models did not increase steadily with increase in number of classes. The 25 class experiment produced a smaller accuracy difference compared to 10 class experiment, while the 50-class experiment had the highest difference in performance between the models. However, later experiments also introduced different species, changed the class distribution and increased the number of available recordings. These factors changed together, so the reduction in accuracy cannot be attributed solely to the number of classes.

The largest macro-F1 difference also occurred in the 50 class experiment. ResNet18 achieved 77.83% compared to 75.81 for ResNet50 giving a difference of 2.02 percentage points. For both the models in this experiment macro-F1 was lower than the weighted-F1. This shows that weighting classes by their validation support produced a higher average compared to giving same importance to every class.

4.2.2 Paired Prediction Analysis

The overall scores show how often each model was correct but they do not show whether the model succeeded and failed on the same audio recordings. Therefore, it is important to do a paired prediction test. For this test the paired predictions were divided into following four outcomes:

- Both models are correct
- Only ResNet18 correct
- Only ResNet50 correct
- Both models incorrect

Table 4.3 presents these test outcomes along with the McNemar test results.

Table 4.3: Paired correctness outcomes for the BirdCLEF validation recordings.

Classes	Both correct	ResNet18 only correct	ResNet50 only correct	Both incorrect	McNemar p
10	817	51	44	88	0.538
25	1,953	127	116	230	0.521
30	2,005	194	179	323	0.469
50	2,505	268	212	533	0.012

Both models correctly classified most of the same validation recordings in every experiment. Their accuracy differences came from the smaller group of recordings for which only one model was correct. For example, in the 30-class test ResNet18 was alone correct for on 194 recordings, while ResNet50 was correct on 179. The net difference therefore was 15 more correct recordings out of 2701 recordings.

The discordant counts were relatively close in the 10-, 25- and 30-class tasks. McNemar’s test did not detect a statistically significant difference for any of these comparisons at the 0.05 threshold. The higher accuracy of ResNet18 in these tasks should therefore be described as an observed advantage rather than statistically established superiority.

The 50-class task produced a larger difference between the discordant counts. ResNet18 correctly classified 268 recordings that ResNet50 classified incorrectly, compared with 212 recordings for which the reverse occurred. This gave a net advantage of 56 correct predictions. The paired test was statistically significant at the unadjusted 0.05 threshold, with $\chi^2 = 6.302$ and $p = 0.012$, providing evidence favouring ResNet18 for this validation split.

The number of recordings classified incorrectly by both models also increased across the tasks. Expressed as a proportion of each validation set, these represented approximately 8.8%, 9.5%, 12.0% and 15.2% for the 10-, 25-, 30- and 50-class tasks respectively. This identifies a growing proportion of examples that neither saved model classified correctly. It does not establish that the models selected the same incorrect label or failed for the same acoustic reason.

These statistical comparisons remain conditional on the evaluated checkpoints and recordings. A non-significant result does not establish equivalence, and a significant result does not account for variation from retraining the models with other seeds. The results provide evidence about the saved model pairs under the stated protocol.

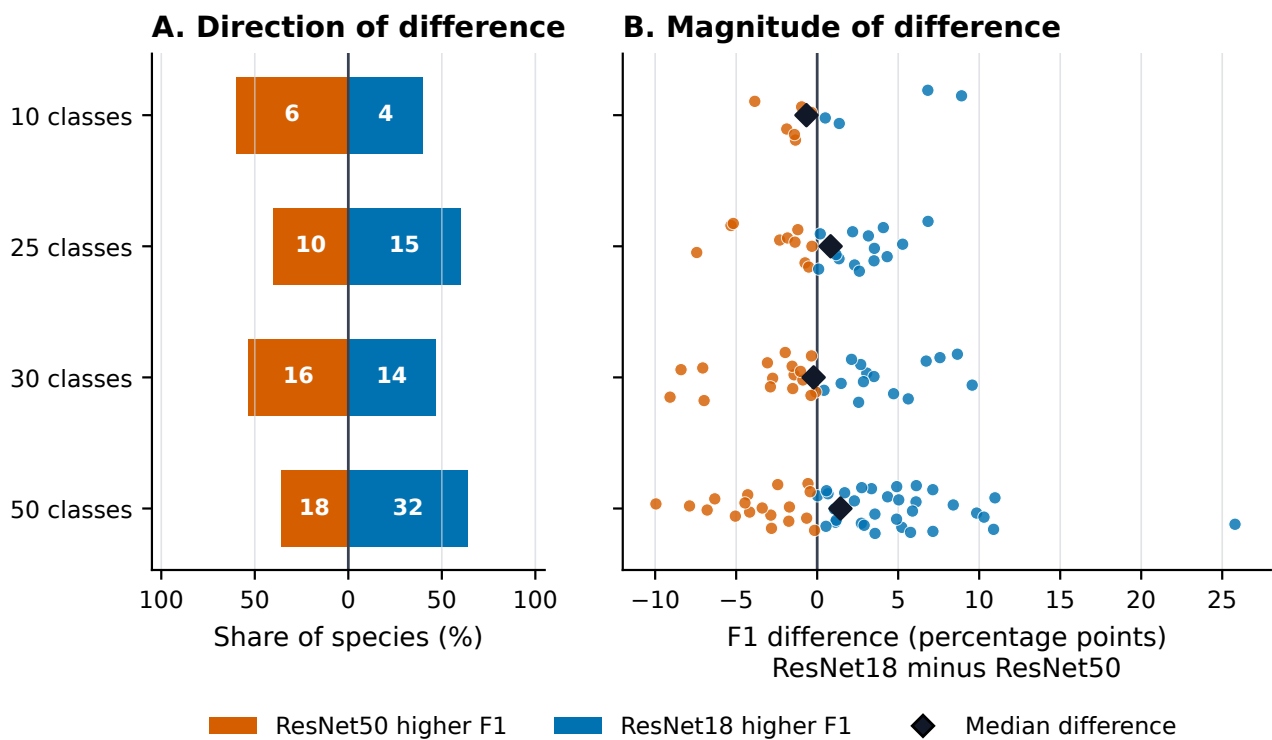
4.2.3 Variation in performance across classes

Aggregate scores can hide the differences between the scores of individual classes. Therefore, per class scores were examined to understand whether the better overall performance by ResNet18 was shared across species or concentrated to a small subset of species.

ResNet18 achieved the higher F1 score for four of the ten species in the 10-class task, 15 of the 25 species, 14 of the 30 species and 32 of the 50 species. There were no exact ties in the saved per-class F1 values. ResNet50 therefore achieved higher F1 for more species in the 10- and 30-class tasks, despite having the lower overall macro-F1 in both cases.

This is possible because macro-F1 depends on the magnitude of the class-level differences, not simply the number of classes favouring each model. A model can obtain a higher macro-F1 through relatively large improvements for a smaller number of species, even if its performance is slightly lower for several others.

Figure 4.2.3 Species-level comparison across BirdCLEF experiments



In the 10-class task, ResNet18 achieved an F1 score of 86.91% for Eurasian Moorhen, compared with 78.00% for ResNet50. It also achieved 86.83% for Blyth's Reed Warbler, compared with 80.00% for ResNet50. These improvements contributed to its higher macro-F1, even though ResNet50 achieved higher F1 for six of the ten species. For Common Rosefinch, for example, ResNet50 achieved 92.86%, compared with 89.01% for ResNet18.

The pattern differed in the 25-class task. ResNet18 achieved higher F1 for a majority of species, including Gray Heron and Black-crowned Night-Heron. However, ResNet50 achieved higher F1 for Eurasian Moorhen and Blyth's Reed Warbler, reversing the direction observed for these species in the 10-class task. Since the class sets were split separately, these comparisons do not necessarily involve the same validation recordings. The reversal therefore cannot be attributed solely to the addition of more classes.

The 30-class task also showed substantial differences in both directions. ResNet18 achieved an F1 score of 75.00% for Little Egret, compared with 65.43% for ResNet50. For Asian Koel, the opposite pattern occurred: ResNet50 achieved 88.07%, compared with 78.99% for ResNet18. These class-level differences were much larger than the overall macro-F1 difference of 0.41 percentage points.

In the 50-class task, the advantage of ResNet18 was distributed across a larger proportion of species, although some classes still favoured ResNet50. The largest positive F1 difference was for Common Myna, where ResNet18 achieved 75.79% and ResNet50 achieved 50.00%. ResNet18 also performed better for Black Kite, with F1 scores of 84.09% and 73.12% respectively. Conversely, ResNet50 achieved higher F1 for Garganey and Common Sandpiper.

The size of a class-level difference must be considered alongside the number of validation examples. Common Myna had 47 validation recordings, while Garganey had 33. With relatively small class supports, changes in a limited number of predictions can noticeably affect precision, recall and F1. The observed differences are useful diagnostic findings, but additional runs would be needed to determine whether they represent stable species-specific advantages.

Overall, the class-level results show that neither architecture was consistently better for every species. Differences in vocalisations, background sounds and recording conditions could contribute to the observed variation, but the present analysis does not isolate these explanations. The evidence supports describing the aggregate ResNet18 advantage as a combination of different class-level outcomes.

4.2.4 Training Behaviour and Checkpoint Selection

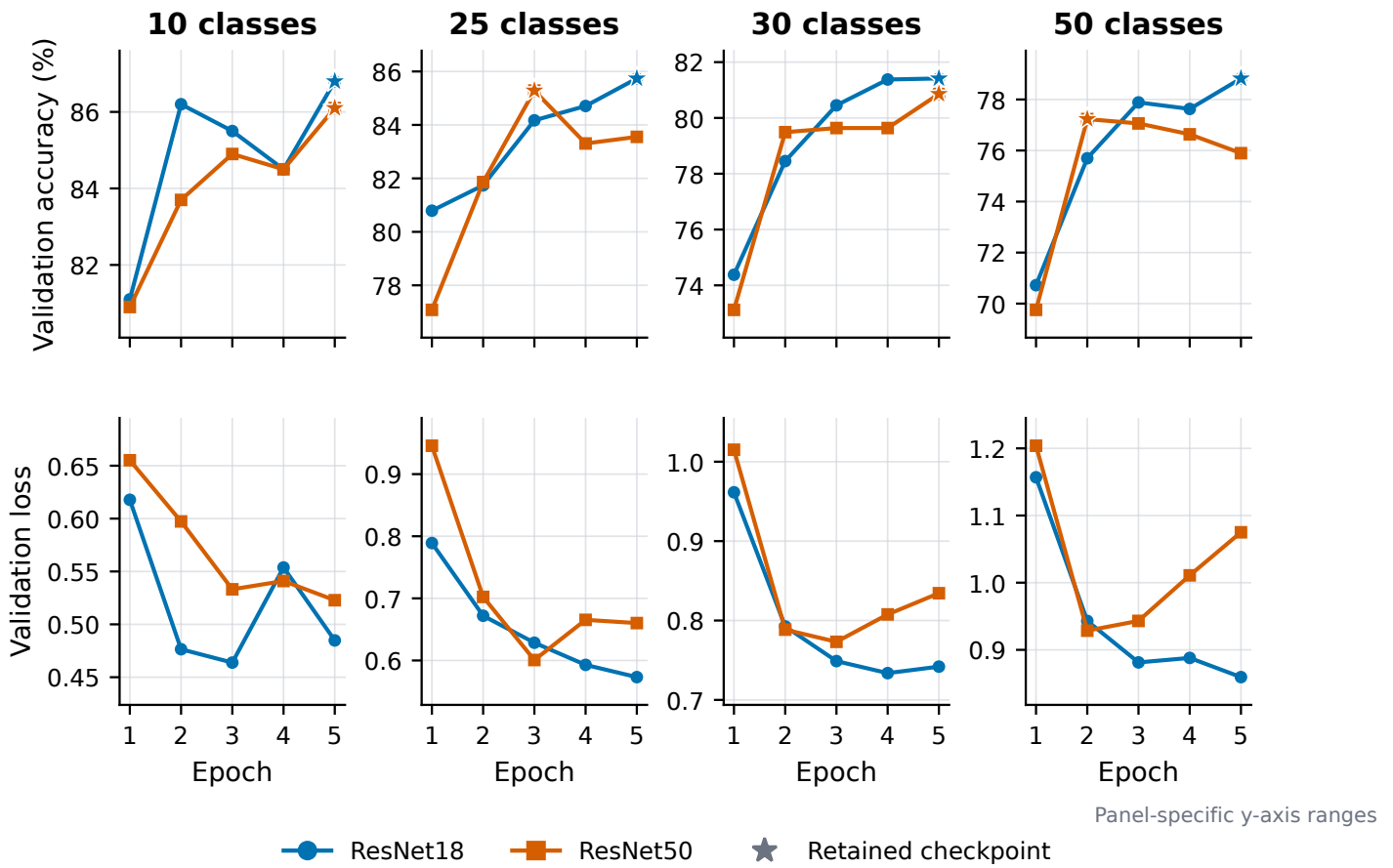
Analysis of training history at each epoch level was saved to understand how the final reported results were developed during the five training epochs. The checkpoint with the highest validation accuracy was saved retained of each training run of the model. ResNet18's retained checkpoint occurred at epoch 5 in all four BirdCLEF tasks. For ResNet50, the retained checkpoints occurred at epochs 5, 3, 5 and 2 for the 10-, 25-, 30- and 50-class tasks respectively.

In the 10-class task, both models achieved their highest validation accuracy at epoch 5. However, ResNet18's validation loss was lowest at epoch 3. The checkpoint selected using accuracy was therefore not the checkpoint that minimised validation loss. This illustrates why the selection criterion must be stated explicitly when reporting the final model.

In the 25-class task, ResNet18's validation accuracy increased throughout training and reached 85.74% at epoch 5. ResNet50 reached its highest validation accuracy of 85.28% at epoch 3, after which its accuracy declined and its validation loss increased. Its training performance nevertheless continued to improve. This pattern is consistent with increasing fit to the training data without a corresponding improvement on the validation set during the later epochs.

Both models achieved their highest validation accuracy at epoch 5 in the 30-class task. Their validation losses reached their minima earlier: epoch 4 for ResNet18 and epoch 3 for ResNet50. Accuracy and cross-entropy loss measure different aspects of the predictions. Accuracy depends on which class receives the highest score, whereas cross-entropy also depends on the probability assigned to the correct class. Consequently, validation accuracy can improve even while the loss increases. The curves alone do not identify which individual predictions caused this behaviour.

Figure 4.2.4: Validation accuracy and loss across five training epochs for the four BirdCLEF tasks. Stars on the accuracy curves identify the checkpoints retained for the final comparison. The corresponding validation losses are shown to distinguish improvements in classification accuracy from changes in the loss.



The 50-class task showed the clearest difference in checkpoint timing. ResNet18 reached its highest validation accuracy at epoch 5, while ResNet50 peaked at epoch 2. ResNet50's validation loss also reached its minimum at epoch 2 and increased during the remaining epochs, despite continued improvement in training performance. Retaining the epoch-2 checkpoint prevented this later decline from determining its reported result.

These observations show that the effect of additional epochs differed between architectures and tasks. They do not support assuming that ResNet50 would necessarily improve simply by continuing the same training procedure. At the same time, five epochs are insufficient to establish the maximum performance achievable by either model under a longer or differently tuned schedule.

4.3 Computational Efficiency and Practical Trade-Off

This experiment examined the computational consequences of the primary 30-class BirdCLEF comparison. The analysis considers predictive performance alongside model size and inference cost. Both models were evaluated using the same input representation and inference procedure.

Table 4.3: Computational and predictive comparison for the BirdCLEF 30-class experiment

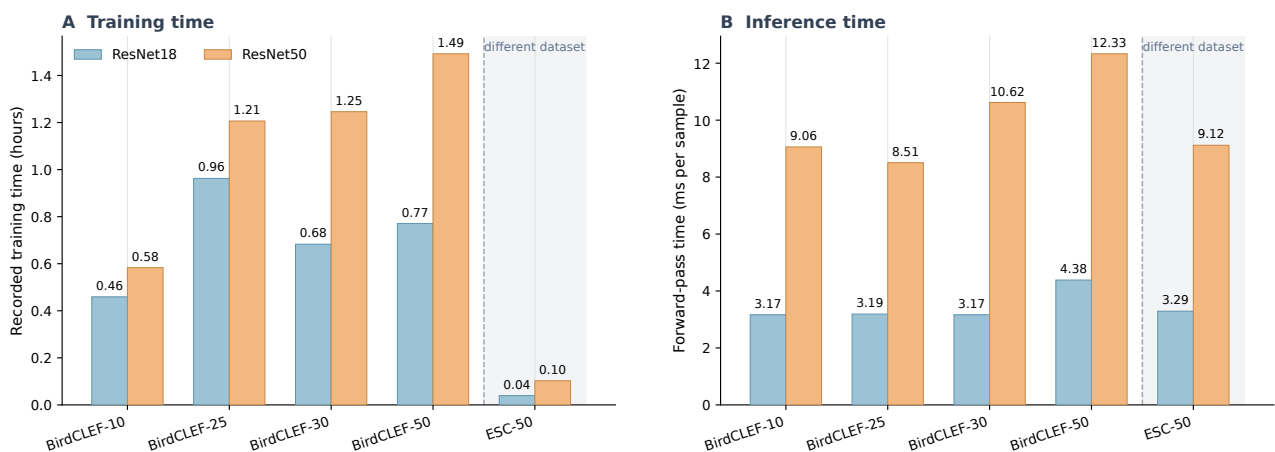
Measure	ResNet18	ResNet50	Relative comparison
Parameters (M)	11.19	23.57	ResNet50 is approximately 2.1 times larger
Inference time (ms/sample)	3.17	10.62	ResNet50 is approximately 3.4 times slower
Validation accuracy (%)	81.41	80.86	ResNet18 is 0.55 percentage points higher
Macro-F1 (%)	81.01	80.60	ResNet18 is 0.41 percentage points higher
Weighted-F1 (%)	81.48	81.08	ResNet18 is 0.40 percentage points higher

ResNet50 contained approximately 23.57 million parameters, compared with 11.19 million for ResNet18. Therefore, ResNet50 was approximately 2.1 times larger in terms of parameter count. The larger model also required more time during inference, with an average of 10.62 milliseconds per sample compared with 3.17 milliseconds for ResNet18. This made ResNet50 approximately 3.4 times slower under the measured inference procedure.

In terms of predictive performance, ResNet18 achieved slightly higher validation accuracy, macro-F1 and weighted-F1 scores. However, the accuracy difference in the 30-class experiment was small and was not statistically significant according to the paired McNemar test. Therefore, the computational results should not be interpreted as showing that ResNet18 was decisively more accurate. Instead, they indicate that ResNet18 achieved comparable predictive performance with substantially lower parameter count and inference cost.

Overall, ResNet18 provided the more favourable practical trade-off for this experiment. It required approximately half the number of parameters and considerably less inference time while producing slightly higher validation scores. These timing measurements are specific to the hardware, software environment, batch configuration and inference procedure used in this study. They should therefore be treated as measurements from this experiment rather than universal hardware-independent values.

Figure 4.3: Training time (A) and inference time (B) for ResNet18 and ResNet50



4.4 Validation performance on ESC-50 dataset

ESC-50 was used as a secondary dataset to examine whether the relative performance of ResNet18 and ResNet50 observed on BirdCLEF generalised to environmental sound classification. Unlike BirdCLEF, which focuses on bird-species recognition, ESC-50 contains 2,000 environmental audio recordings distributed across 50 sound classes, with 40 recordings per class.

The experiment used the prepared stratified 80:20 split with random seed 42. This produced 1,600 training recordings and 400 validation recordings. The same audio preprocessing pipeline was applied as in the BirdCLEF experiments: recordings were converted to mono, resampled to 32 kHz, standardised to five seconds, and represented as 128-bin mel-spectrograms. Both models were initialized with ImageNet-pretrained weights, trained using the Adam optimizer with a learning rate of 0.0001, a batch size of 32, and a five-epoch training budget.

Table 5.5: BirdCLEF 50-class validation performance

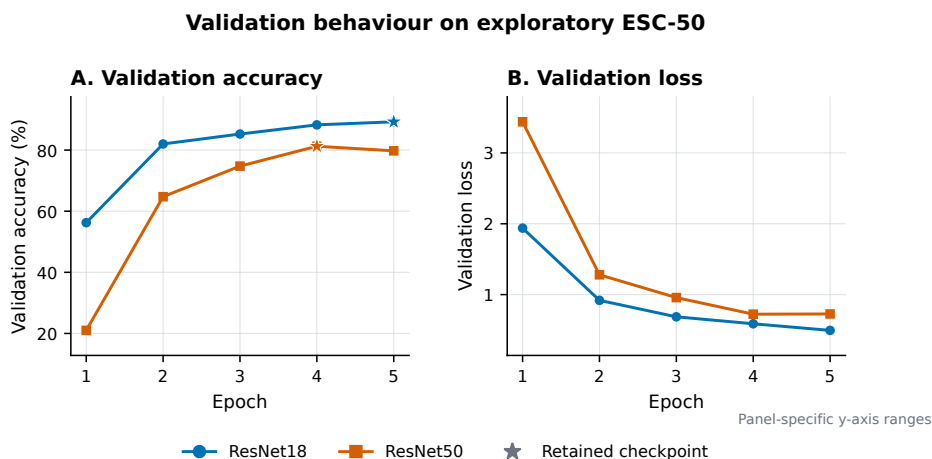
Model	Accuracy	95% CI	Macro-F1	Weighted-F1
ResNet18	89.25%	[85.99, 92.00]	89.41%	89.41%
ResNet50	81.25%	[77.50, 85.00]	80.42%	80.42%

ResNet18 achieved a validation accuracy of 89.25% and a macro-F1 score of 89.41%, while ResNet50 achieved an accuracy of 81.25% and a macro-F1 score of 80.42%. ResNet18 therefore achieved an accuracy advantage of 8.00 percentage points and a macro-F1 advantage of 8.99 percentage points.

The bootstrap 95% confidence intervals for accuracy were [85.99, 92.00] for ResNet18 and [77.50, 85.00] for ResNet50. The paired McNemar test produced $\chi^2 = 13.729$ and $p = 0.000211$. ResNet18 was correct while ResNet50 was incorrect on 51 validation recordings, whereas ResNet50 was correct while ResNet18 was incorrect on 19 recordings. The paired comparison was therefore statistically significant for this validation split.

These results provide supporting evidence that ResNet18 performed better than ResNet50 on this environmental sound classification task. However, the result should be interpreted cautiously. The experiment used a random stratified 80:20 split rather than ESC-50's five official source-aware folds, and 129 source files overlapped between the training and validation partitions. Therefore, this experiment provides exploratory supporting evidence rather than a definitive estimate of generalisation performance.

Figure 4.4 Validation loss and accuracy curves for ResNet18 and ResNet50



4.5 Findings and Discussion

This chapter compared ResNet18 and ResNet50 across four BirdCLEF tasks, an exploratory ESC-50 experiment and an analysis of their computational requirements. The findings are discussed below in relation to the four research questions and the previous literature.

For RQ1, ResNet18 achieved higher validation accuracy, macro-F1 and weighted-F1 than ResNet50 in all four BirdCLEF tasks. The differences were relatively small, ranging from 0.46 to 1.59 percentage points in accuracy. The paired McNemar tests found a statistically significant difference only in the 50-class task. Therefore, the results show a consistent observed advantage for ResNet18, but they do not establish that it was significantly better in every task. Additional runs with different random seeds would be required to determine whether these differences are stable.

One possible explanation is that ResNet18 was better suited to the short five-epoch training protocol used in this study. ResNet50 has more layers and parameters, which may require a longer or differently tuned training process to make effective use of its additional capacity. This interpretation is supported by the checkpoint results. ResNet18 achieved its best validation accuracy at epoch five in every BirdCLEF task, while ResNet50 reached its best checkpoint earlier in the 25- and 50-class tasks. However, the experiment does not prove that additional training would improve ResNet50.

For RQ2, the overall pattern remained similar as the number of classes increased. ResNet18 obtained higher overall scores in the 10-, 25-, 30- and 50-class tasks, with the largest difference occurring in the 50-class experiment. However, the difference did not increase steadily with the number of classes because the species, number of recordings and class distributions also changed between experiments. The reduction in accuracy cannot therefore be attributed only to increasing the number of classes.

The class-level analysis also showed that ResNet18 was not better for every species. ResNet50 achieved higher F1 scores for some species even when its overall macro-F1 was lower. This suggests that the architectures may respond differently to particular vocalisations, background sounds and recording conditions. However, some classes contained relatively few validation recordings, so changes in a small number of predictions could produce noticeable differences in their F1 scores. Repeated experiments would be needed before concluding that either architecture is more suitable for particular species.

For RQ3, the exploratory ESC-50 experiment also favoured ResNet18. Its validation accuracy was 89.25%, compared with 81.25% for ResNet50, and the paired difference was statistically significant. This provides some evidence that the advantage of ResNet18 was not limited to bird-species classification. However, the random split allowed recordings from the same source files to appear in both the training and validation sets. The ESC-50 result may therefore overestimate performance on recordings from completely unseen sources and should be treated as supporting evidence rather than a definitive comparison.

For RQ4, ResNet18 provided the better balance between predictive performance and computational cost. In the primary 30-class experiment, it achieved slightly higher validation scores while using approximately half as many parameters as ResNet50. Its measured inference time was also approximately 3.4 times faster. This result is practically relevant because a smaller and faster model would be easier to use on devices with limited memory or processing power.

These results are consistent with previous work showing that increasing the depth of a residual network does not always produce better audio-classification performance (Kong et al., 2020). However, they differ from Theivaprakasham et al. (2023), who reported stronger performance from

ResNet50 for bird-audio identification. Their models were trained using a much longer fine-tuning procedure and different data and preprocessing settings. The difference between the studies suggests that model depth alone does not determine performance; the training duration, dataset, preprocessing and optimisation procedure also influence the result.

Overall, ResNet18 was the stronger practical choice under the conditions tested in this study. It produced slightly better BirdCLEF results, performed better in the exploratory ESC-50 comparison and required fewer computational resources. This conclusion is limited to the selected datasets, data splits, saved checkpoints and five-epoch training protocol. It should not be interpreted as evidence that ResNet18 will always outperform ResNet50. Instead, the findings show that the additional depth and computational cost of ResNet50 did not provide an advantage under the controlled conditions used in this dissertation.

5. Future Work

This study compared ResNet18 and ResNet50 for audio classification. Further experiments could help us understand whether the results remain similar under different training conditions and with different datasets.

One extension would be to train both models several times using different random seeds. Comparing the average results would help show whether ResNet18 performs better consistently or whether the results change between runs.

Another improvement would be to keep recordings from the same source together when dividing the data into training and validation sets. For ESC-50, the five official folds could be used. This would give a clearer idea of how well the models perform on recordings from new sources.

Both models could also be trained with modified audio, such as adding background noise or slightly shifting the sound within a recording. This would help test whether a wider variety of training examples improves their performance.

Further experiments could use more recordings and more bird species. Training both models for more epochs would also help show whether ResNet50 benefits from additional data or training time.

Finally, transformer-based audio models could be compared with the two ResNet models. Using similar data and training conditions would help show how their accuracy and computational requirements compare.

Appendix

This appendix records the values used in the added interpretation so that the narrative can be checked against the project outputs. It is not a replacement for the saved CSV files or model checkpoints. Its purpose is to make the dissertation’s result provenance visible and to prevent values from separate class-set experiments being mixed together.

BirdCLEF 10-class run: ResNet18 accuracy 86.80%, macro-F1 86.85%, weighted-F1 86.85% and 11.18 million parameters; ResNet50 accuracy 86.10%, macro-F1 86.07%, weighted-F1 86.07% and 23.53 million parameters. McNemar $\chi^2 = 0.379$, $p = 0.5382$, with discordant counts 51 in favour of ResNet18 and 44 in favour of ResNet50.

BirdCLEF 25-class run: ResNet18 accuracy 85.74%, macro-F1 85.97% and weighted-F1 85.90%; ResNet50 accuracy 85.28%, macro-F1 85.37% and weighted-F1 85.35%. McNemar $\chi^2 = 0.412$, $p = 0.5212$, with discordant counts 127 and 116. The point estimate favours ResNet18 but the paired difference is not significant.

BirdCLEF 30-class run: The validation set contained 2,701 recordings. ResNet18 achieved an accuracy of 81.41% (95% CI [79.97%, 82.86%]), a macro-F1 of 81.01% and a weighted-F1 of 81.48%. ResNet50 achieved an accuracy of 80.86% (95% CI [79.45%, 82.34%]), a macro-F1 of 80.60% and a weighted-F1 of 81.08%. McNemar’s test was not statistically significant ($\chi^2 = 0.525$, $p = 0.469$), with discordant counts of 194 in favour of ResNet18 and 179 in favour of ResNet50.

BirdCLEF 50-class run: ResNet18 accuracy 78.82%, 95% CI [77.49%, 80.16%], macro-F1 77.83% and weighted-F1 78.84%. ResNet50 accuracy 77.23%, 95% CI [75.92%, 78.57%], macro-F1 75.81% and weighted-F1 77.27%. McNemar $\chi^2 = 6.302$, $p = 0.0121$, with discordant counts 268 and 212.

ESC-50 exploratory random-split run: ResNet18 accuracy 89.25% with 95% CI [85.99%, 92.00%] and macro-F1 89.41%; ResNet50 accuracy 81.25% with 95% CI [77.50%, 85.00%] and macro-F1 80.42%. McNemar $\chi^2 = 13.729$, $p = 0.000211$, with discordant counts 51 and 19. This result carries the source-overlap limitation described in the limitations section.

The primary saved result sources are the `summary_comparison.csv` and `mcnemar_results.csv` files in the project `comparison_results` folders for the 10-, 25-, 30- and 50-class BirdCLEF runs and the ESC-50 experiment folder. Where absolute inference times differ across runs, the values should remain attached to their original run and measurement environment rather than averaged as if they were hardware-independent.

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